

Unraveling Glioblastoma: TME Implication and Gene Therapy Advances.

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Glioblastoma is a malignant manifestation of a solid brain tumour with a very dismal prognosis due to an overall median survival of 14 months. The currently administered Standard treatment plan, the STUPP regimen, is not very effective in tackling this neoplasia. A major concern that affects the development of new drug formulations, specifically for Glioma, is the inherent sub-clonal heterogeneity, which includes the dynamic and intricate nature of the Tumour Microenvironment (TME). Targeting the cellular niche using personalized medication for glioma specifically gene therapy, seems to be promising, with most studies in preclinical models yielding optimistic results. This paper analyses the great headways made in glioma gene therapy in the last 10 years while looking into different therapeutic strategies. That said, certain challenges do plague the clinical use of gene therapy which have been highlighted in the hopes that future researchers will address these concerns and further propel gene therapy in its journey from the Lab to the bedside.

Longitudinal multimodal profiling of IDH-wildtype glioblastoma reveals the molecular evolution and cellular phenotypes underlying prognostically different treatment responses.

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Despite recent advances in the biology of IDH-wildtype glioblastoma, it remains a devastating disease with median survival of less than 2 years. However, the molecular underpinnings of the heterogeneous response to the current standard-of-care treatment regimen consisting of maximal safe resection, adjuvant radiation, and chemotherapy with temozolomide remain unknown. Comprehensive histopathologic, genomic, and epigenomic evaluation of paired initial and recurrent glioblastoma specimens from 106 patients was performed to investigate the molecular evolution and cellular phenotypes underlying differential treatment responses. While TERT promoter mutation and CDKN2A homozygous deletion were early events during gliomagenesis shared by initial and recurrent tumors, most other recurrent genetic alterations (eg, EGFR, PTEN, and NF1) were commonly private to initial or recurrent tumors indicating acquisition later during clonal evolution. Furthermore, glioblastomas exhibited heterogeneous epigenomic evolution with subsets becoming more globally hypermethylated, hypomethylated, or remaining stable. Glioblastoma that underwent sarcomatous transformation had shorter interval to recurrence and were significantly enriched in NF1, TP53, and RB1 alterations and the mesenchymal epigenetic class. Patients who developed somatic hypermutation following temozolomide treatment had significantly longer interval to disease recurrence and prolonged overall survival, and increased methylation at 4 specific CpG sites in the promoter region of MGMT was significantly associated with this development of hypermutation. Finally, an epigenomic evolution signature incorporating change in DNA methylation levels across 347 critical CpG sites was developed that significantly correlated with clinical outcomes. Glioblastoma undergoes heterogeneous genetic, epigenetic, and cellular evolution that underlies prognostically different treatment responses.

Research progress of drug resistance mechanism of temozolomide in the treatment of glioblastoma.

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Glioblastoma, the most malignant primary brain tumor among gliomas, is characterized by a low cure rate, high recurrence rate, and invasive growth. Without chemotherapy, the median survival of patients is only 12.1 months. The standard treatment for glioblastoma primarily involves surgical resection, complemented by radiotherapy. Temozolomide (TMZ), a new oral alkylating agent, is currently used as the first-line chemotherapy drug for glioma. However, TMZ treatment only improves median survival by 2 months, largely because of the tumor's ability to develop resistance to the drug. The main mechanism underlying this resistance involves DNA repair processes, such as the action of O6-methylguanine DNA methyltransferase (MGMT), which repairs the DNA damage caused by TMZ, and other DNA repair mechanisms including mismatch repair and base excision repair. These mechanisms can effectively repair the DNA damage caused by TMZ, thereby reducing the sensitivity of tumor cells to the drug. This study summarized the recent research progress of TMZ resistance mechanism in glioblastoma, aiming to provide a theoretical basis for the development of new therapies. The mechanisms of glioma resistance to TMZ mainly involves DNA damage repair (as mentioned above), abnormal cell signaling pathways (p53-mediated signaling, reactive oxygen species-mediated signaling, endoplasmic reticulum stress and autophagy-related signaling, receptor tyrosine kinase-related signaling, transforming growth factors, β -mediated signaling pathway, Wnt/ β -Catenin signaling pathway), glioma stem cells, tumor microenvironment (hypoxic microenvironment, nano-drug delivery system), epidermal growth factor receptor, and microRNAs.

Effective Treatment of Glioblastoma Multiforme With Oncolytic Virotherapy: A Case-Series.

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Glioblastoma multiforme (GBM) remains an incurable condition, associated with a median survival time of 15 months with best standard of care and 5-year survival rate of <10%. We report on four GBM patients on combination treatment regimens that included oncolytic virus (OV) immunotherapy, who achieved clinical and radiological responses with long-term survival, thus far, of up to 14 years, and good quality of life. We discuss the radiological findings that provide new insights into this treatment, the scientific rationale of this innovative and promising therapy, and considerations for future research.

Advanced tumor electric fields therapy: A review of innovative research and development and prospect of application in glioblastoma.

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Glioblastoma multiforme (GBM) is an aggressive malignant tumor with a high mortality rate and is the most prevalent primary intracranial tumor that remains incurable. The current standard treatment, which involves surgery along with concurrent radiotherapy and chemotherapy, only yields a survival time of 14-16 months. However, the introduction of tumor electric fields therapy (TEFT) has provided a glimmer of hope for patients with newly diagnosed and recurrent GBM, as it has been shown to extend the median survival time to 20 months. The combination of TEFT and other advanced therapies is a promising trend in the field of GBM, facilitated by advancements in medical technology. In this review, we provide a concise overview of the mechanism and efficacy of TEFT. In addition, we mainly discussed the innovation

of TEFT and our proposed blueprint for TEFT implementation. Tumor electric fields therapy is an effective and highly promising treatment modality for GBM. The full therapeutic potential of TEFT can be exploited by combined with other innovative technologies and treatments.

Looking for the Holy Grail-Drug Candidates for Glioblastoma Multiforme Chemotherapy.

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Glioblastoma multiforme (GBM) is the deadliest and the most heterogeneous brain cancer. The median survival time of GBM patients is approximately 8 to 15 months after initial diagnosis. GBM development is determined by numerous signaling pathways and is considered one of the most challenging and complicated-to-treat cancer types. Standard GBM therapy consist of surgery followed by radiotherapy or chemotherapy, and combined treatment. Current standard of care (SOC) does not offer a significant chance for GBM patients to combat cancer, and the selection of available drugs is limited. For almost 20 years, there has been only one drug, Temozolomide (TMZ), approved as a first-line GBM treatment. Due to the limited efficacy of TMZ and the high rate of resistant patients, the implementation of new chemotherapeutics is highly desired. However, due to the unique properties of GBM, many challenges still need to be overcome before reaching a 'breakthrough'. This review article describes the most recent compounds introduced into clinical trials as drug candidates for GBM chemotherapy.

Breaching barriers in glioblastoma. Part I: Molecular pathways and novel treatment approaches.

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Glioblastoma multiforme (GBM) is the most common primary brain tumour, and the most aggressive in nature. The prognosis for patients with GBM remains poor, with a median survival time of only 1-2 years. The treatment failure relies on the development of resistance by tumour cells and the difficulty of ensuring that drugs effectively cross the dual blood brain barrier/blood brain tumour barrier. The advanced molecular and genetic knowledge has allowed to identify the mechanisms responsible for temozolomide resistance, which represents the standard of care in GBM, along with surgical resection and radiotherapy. Such resistance has motivated the researchers to investigate new avenues for GBM treatment intended to improve patient survival. In this review, we provide an overview of major obstacles to effective treatment of GBM, encompassing biological barriers, cancer stem cells, DNA repair mechanisms, deregulated signalling pathways and autophagy. New insights and potential therapy approaches for GBM are also discussed, emphasizing localized chemotherapy delivered directly to the brain, immunotherapy, gene therapy and nanoparticle-mediated brain drug delivery.

CDK4/6 Inhibitor PD0332991 in Glioblastoma Treatment: Does It Have a Future?

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Glioblastoma is aggressive, highly infiltrating, and the most frequent malignant form of brain cancer. With a median survival time of only 14.6 months, when treated with the standard of care, it is essential to

find new therapeutic options. A specific CDK4/6 inhibitor, PD0332991, obtained accelerated approval from the Food and Drug Administration for the treatment of patients with advanced estrogen receptor-positive and HER2-negative breast cancer. Common alterations in the cyclin D1-cyclin-dependent kinase 4/6-retinoblastoma 1 pathway in glioblastoma make PD0332991 also an interesting drug for the treatment of glioblastoma. Promising results in in vitro studies, where patient derived glioblastoma cell lines showed sensitivity to PD0332991, gave motive to start in vivo studies. Outcomes of these studies have been contrasting in terms of PD0332991 efficacy within the brain: more research is necessary to conclude whether CDK4/6 inhibitor can be beneficial in the treatment of glioblastoma.

RNA Sequencing in Hypoxia-Adapted T98G Glioblastoma Cells Provides Supportive Evidence for IRE1 as a Potential Therapeutic Target.

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Glioblastoma (GBM) is an aggressive brain cancer with a median survival time of 14.6 months after diagnosis. GBM cells have altered metabolism and exhibit the Warburg effect, preferentially producing lactate under aerobic conditions. After standard-of-care treatment for GBM, there is an almost 100% recurrence rate. Hypoxia-adapted, treatment-resistant GBM stem-like cells are thought to drive this high recurrence rate. We used human T98G GBM cells as a model to identify differential gene expression induced by hypoxia and to search for potential therapeutic targets of hypoxia adapted GBM cells. RNA sequencing (RNAseq) and bioinformatics were used to identify differentially expressed genes (DEGs) and cellular pathways affected by hypoxia. We also examined expression of lactate dehydrogenase (LDH) genes using qRT-PCR and zymography as LDH dysregulation is a feature of many cancers. We found 2630 DEGs significantly altered by hypoxia ($p < 0.05$), 1241 upregulated in hypoxia and 1389 upregulated in normoxia. Hypoxia DEGs were highest in pathways related to glycolysis, hypoxia response, cell adhesion and notably the endoplasmic reticulum, including the inositol-requiring enzyme 1 (IRE1)-mediated unfolded protein response (UPR). These results, paired with numerous published preclinical data, provide additional evidence that inhibition of the IRE1-mediated UPR may have therapeutic potential in treating GBM. We propose a possible drug repurposing strategy to simultaneously target IRE1 and the spleen tyrosine kinase (SYK) in patients with GBM.

Neoadjuvant gemcitabine/treosulfan chemotherapy for newly diagnosed glioblastoma: a phase II study.

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The median survival for patients with glioblastoma is 12 months. The authors evaluated whether preirradiation gemcitabine/treosulfan (GeT) chemotherapy followed by standard radiotherapy improved outcome in patients with glioblastoma. Seventeen patients with newly diagnosed glioblastoma were enrolled in a prospective, unicenter trial of preirradiation GeT chemotherapy. Chemotherapy included up to 4 cycles of intravenous gemcitabine (1000 mg/m² body surface) and treosulfan (3500 mg/m² body surface) on days 1 and 8 of 28 days treatment cycles. Involved field radiotherapy (60 Gy in 30 fractions) was given after chemotherapy or earlier in the case of disease progression or drug intolerance. There was no specific treatment-related neurotoxicity reported, but in 3 of 17 patients (18%) chemotherapy

was stopped because of World Health Organization (WHO) IV hematological toxicity. With GeT chemotherapy alone, there was a median progression-free survival of 12 weeks and a progression-free survival rate at 4 months of 29%. In 16 of 17 patients who subsequently received a full course of radiotherapy, the median progression-free survival from the time of diagnosis was 8 months, and the progression-free survival rate at 12 months was 25% (4 of 16 patients). The median overall survival was 12 months. Neither age nor extent of the residual postoperative tumor predicted the duration of progression-free survival after chemotherapy alone or after chemotherapy followed by radiotherapy. The combination of gemcitabine and treosulfan produced significant hematological toxicity in patients with newly diagnosed glioblastoma. The schedule used in the present study did not confer any significant survival advantage compared with standard involved field radiotherapy alone.
